AI-Driven Decision Support Framework for Intelligent Quick Commerce Operations

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***Abstract*—Quick commerce (Q-commerce) delivers groceries and everyday essentials within ten to thirty minutes from networks of small urban warehouses called dark stores. Keeping that promise requires thousands of linked decisions per store per hour on stocking, reordering, order-to-store assignment, batching and dispatch. These are taken under demand that is sparse for any one product at one store in one time slot, over stock that expires, and at profit margins now measured in low single digits of order value. In practice the supporting analytics are fragmented: forecasting, inventory, dispatch and routing are optimised separately, models are trained for forecast accuracy rather than for the decisions they drive, outputs are rarely explained to the planners who handle exceptions, and the customer reviews that feed assortment and new-product decisions are used without any check on authenticity. This paper proposes a unified decision support framework in five layers (data, features, models, decision support and human-in-the-loop) that connects demand forecasting, product recommendation, delivery-time estimation and routing, inventory optimisation and a proposed review-authenticity check through three explicit couplings and one shared cost score. Forecasting models are trained on the cost of the decisions they lead to rather than on forecast accuracy. The problem is grounded in operating metrics disclosed by Indian platforms, three of which publish the figures used here. We set out an evaluation plan over public datasets and a dark-store simulator, state five testable hypotheses, and report a small simulated experiment in which training for the decision lowers inventory cost by 10.3% while scoring worse on forecast accuracy, the direction the framework predicts.**

Keywords*—*quick commerce, dark stores, decision support systems, demand forecasting, inventory optimisation, last-mile delivery, fake review detection, Indian retail

I. INTRODUCTION

Quick commerce is the retail model in which a limited range of grocery, personal-care and household items reaches an urban customer within about ten to thirty minutes of ordering. Conventional e-grocery consolidates demand in large regional warehouses, plans picking and routing hours ahead and delivers in multi-hour slots. Q-commerce instead spreads stock across many dark stores of roughly 2,000 to 5,000 square feet, each drawing on a city-wide catalogue (about 20,000 products in smaller cities and close to 80,000 across a metro such as Delhi NCR [1]) but carrying only a selected subset, a few units deep per product, located so that most customers are within two to three kilometres. It commits to a delivery time the moment an order is placed, before picking starts, and it plans fulfilment continuously rather than in cycles.

India is the natural place to study these operations. It has the largest and fastest-growing Q-commerce sector outside China, several operators are listed or filing to list and therefore publish quarterly figures, and urban density, traffic and the gig-labour market make the operating constraints unusually tight. Bain and Flipkart estimate gross merchandise value (GMV) at USD 10–11 billion in CY2025, roughly 16–17% of Indian e-retail, and project USD 65–70 billion by 2030 [2]. Anand Rathi puts FY26 GMV near USD 11.3 billion, up from about USD 1.6 billion in FY23, a sevenfold rise in three years [3]; CareEdge reports 142% compound annual growth over FY22–FY25 and projects ₹2 lakh crore by FY28 [4]. By August 2026 the sector was handling 9.1 to 9.5 million orders a day across six platforms [5], with more than 50 million monthly transacting users and an estimated 5,500 to 7,000 dark stores at an average order value near ₹460 [6]. Published market-size figures differ because they define the market differently (revenue versus GMV, with or without non-grocery categories), so we report the range and its sources rather than pick one number.

The operational problem matters because of unit economics. In 2026 the leading operators reached or crossed break-even at single-digit margins. Blinkit reported adjusted EBITDA of +0.6% of net order value (NOV) in Q1 FY27, and Swiggy Instamart reached a contribution margin of −0.2% of gross order value (GOV) in the same quarter, up from a −5.6% low five quarters earlier [7], [8]. The two measures are defined differently and are not comparable; what they share is their size. At such margins a stockout, a spoiled perishable or a late delivery is not noise. Blinkit's disclosed inventory loss of ₹308 crore in one quarter, about 1.8% of NOV, is roughly three times its entire adjusted EBITDA margin [7]. On current economics, waste and availability are the first-order problems.

Six things make Q-commerce operations different from conventional retail. Demand has to be forecast per product, per store, per hour or even per fifteen minutes, where most series are mostly zeros. Stock is shallow (six to twelve units of an item), so small forecast errors become stockouts that are either hidden from the app, corrupting the demand data, or substituted at pick time, costing margin. Delivery runs against a hard, short deadline with uncertain picking times, travel times and rider supply. New products and new stores with no history are a permanent condition, because ranges change quickly and networks add stores by the hundred each quarter. The trade-offs have no single right answer: fewer stockouts mean more waste, tighter delivery promises mean worse batching, and the right balance differs by store and hour. Finally, the trust signal is polluted. Reviews and ratings feed assortment and new-

product decisions, and that channel is measurably degraded: e-commerce grievances filed with India's National Consumer Helpline rose from 95,270 (22% of all grievances) in 2018 to 444,034 (43%) in 2023 [9]; a large consumer survey found that most online shoppers believe ratings are positively biased [10]; the Bureau of Indian Standards has issued IS 19000:2022 on online reviews, voluntary for now with consultation on making it mandatory [11]; and in November 2025 twenty-six platforms filed dark-pattern compliance declarations with the CCPA [12].

The paper asks one question: how can an integrated AI-driven decision support framework improve demand forecasting, inventory optimisation, last-mile delivery, recommendation, operational decision-making and review authenticity in quick commerce, and under what conditions does integrating these functions beat optimising them separately? The second half is the testable part. Section II reviews the literature, Section III presents the method and framework, Section IV describes the data and evaluation plan, Section V reports a first experiment and maps where integration exists and where it does not, and Section VI sets out the limitations.

II. LITERATURE REVIEW

Demand forecasting and inventory. The M5 competition on hierarchical Walmart sales established that gradient-boosted tree models trained across many series at once beat one-model-per-series approaches, and that combining models beats any single one [13]. Three model families remain relevant: classical baselines such as Prophet [14]; gradient-boosted trees, mainly XGBoost [15] and LightGBM [16]; and deep sequence models such as DeepAR [17] and the Temporal Fusion Transformer [18], which output a range of outcomes over several horizons, which is what replenishment needs. Forecasts at product, store, zone and city level are made consistent with each other by reconciliation methods [19], [20]. Rarely-selling items need their own treatment: Croston's method with the Syntetos–Boylan correction is still the standard baseline [21], and pretrained time-series models such as Chronos [22] and TimesFM [23] can forecast a series they have never seen, which is exactly the new-product case. On the inventory side, Ban and Rudin learn ordering rules directly from data [24], and Gijsbrechts et al. find deep reinforcement learning competitive but not clearly better than classical methods, and sensitive to tuning [25].

Training for the decision. The usual pipeline forecasts first and optimises second, treating the forecast as if it were true. Elmachtoub and Grigas show this is generally not the best approach and propose training the forecaster on the cost of the decision it produces [26]; Mandi et al. survey the field [27]. This matters in Q-commerce because forecast errors in the two directions cost very different amounts and the difference varies by product. The decision-support literature adds a warning that research has drifted toward technique and away from the person making the decision [28], which applies to many machine-learning systems deployed with little thought for how a planner will use them.

Last-mile delivery. Vehicle routing and its variants are the formal backbone [29], [30]. The closest problem types are restaurant meal delivery with deadlines [31], dispatching from stores under uncertain demand [32] and courier dispatch [33]. Learned travel-time estimation is in production at Google Maps [34], and Raj and Das study delivery fees, penalties and rider–platform cooperation in Q-commerce specifically [35]. What this literature does not do is treat stock availability as an input: the store is assumed to have what the order needs.

Recommendation. Grocery recommendation is dominated by repeat purchase. Two-stage retrieve-then-rank models are the standard design [36], the Instacart dataset is the main public testbed [37], and repeat-aware next-basket models target grocery directly [38]. Li et al. show that much of the reported gain of complex models disappears against a simple "recommend what they bought last time" baseline once repeat and new items are scored separately [39], so we treat that baseline as compulsory.

Generative and agentic AI. This is the newest and least settled area. Li et al. present a language-model interface that turns questions into changes to a supply chain optimiser [40]; agent designs rest on interleaved reasoning and acting [41] and multi-agent frameworks such as AutoGen [42]. One unrefereed preprint reports that LLMs can grade route quality at 79–86% agreement on a small sample [43]. We could not find peer-reviewed evidence of LLM agents handling supply-chain exceptions in production with measured results, so the framework gives this layer a limited role and treats its value as a hypothesis.

Fake-review detection. The field goes back to Jindal and Liu [44] and to Ott et al., who showed that human judges perform near chance while trained classifiers do better [45]. Rayana and Akoglu showed that combining review text with the reviewer–product graph beats text alone [46], and later graph models were hardened against fraudsters who disguise themselves [47], fused text with graph structure [48], used reinforcement learning to pick informative neighbours [49], and applied contrastive learning to camouflage [50]. Most recently, FraudSquad feeds language-model embeddings of review text into a graph transformer to catch LLM-written spam [51]; it is the closest precedent for the text-plus-graph check in Section III, and it also shows how far the field is from deployment, since it beats baselines by a wide margin on synthetic LLM-spam data but reaches only about 34% precision and 38% recall on human-labelled Yelp data. Separately, detection of AI-generated images has moved from GAN-era classifiers [52] to methods that generalise across image generators [53], [54]. None of the review-detection methods uses image evidence, and there is no benchmark for image authenticity inside review content; Section VI returns to this gap.

III. METHODOLOGY

A. Problem statement

A quick-commerce network makes two kinds of decisions at two speeds. The slow decisions, taken a few times a day, are about inventory: how many units of each product to keep at each dark store, when to reorder, and which products to list where. The fast decisions, taken every few seconds, are about orders: which store should serve an incoming order, which orders should travel together, which rider takes the batch, in what sequence the drops are made, and what to substitute when an item is missing at pick time. The two are connected by one piece

of state, the stock actually available at each store right now, which the slow decisions create and the fast decisions consume.

We measure the network on four things: total operating cost (holding stock, reordering, writing off expired stock, delivery and picking labour); service, meaning the share of orders delivered by the promised time, with the slowest 5% watched separately because a few very late orders lose more customers than a slightly slow average; availability, meaning the share of demanded units that were actually in stock; and rider utilisation. For the optimiser these are combined into a single cost score: operating cost, plus a penalty for every minute an order is late, plus a penalty for every unit of demand lost to a stockout, minus a small credit for rider utilisation. The penalty weights are business choices, not model outputs, and the framework reports how the trade-offs shift as they move rather than claiming one setting is correct. Stock is issued first-expired-first-out, each order goes to exactly one store within delivery range, and routes obey the usual vehicle-routing constraints [29].

B. The three couplings

Three links connect sub-problems that current systems solve separately. They are the core contribution of the paper, and the evaluation in Section IV is designed to switch each one off and measure what is lost.

C1, substitution to inventory. When a product is out of stock, the loss depends on whether the customer will accept a substitute. The recommendation model already estimates that probability. We use it to discount the stockout penalty: if the substitute-acceptance probability for a product at a store is π, the penalty for running out of it is multiplied by $(1 - \pi)$. A product with a well-accepted substitute on the shelf is cheaper to run out of, so the inventory optimiser holds less of it.

C2, availability to assignment. Today an order is routed to the nearest store on the assumption that the store has everything. We instead let projected stock decide which stores are eligible: a store qualifies only if it is within delivery range and is expected to have most of the basket in stock. Only then do distance and rider availability decide. This is the point where inventory state enters routing, which the routing literature normally treats as given.

C3, review trust to new-product forecasts. A new product has no sales history, so its first demand estimate borrows from signals such as reviews and ratings. We weight each review by its estimated authenticity score, a number between 0 and 1 produced by the review-authenticity component, so that suspicious reviews count for less. That score is used only here; it never changes replenishment or routing directly, because there is no evidence to support such a link.

C. Training the forecast for the decision

The usual approach trains a demand forecast to be as close as possible to actual sales and then hands the number to the inventory rule. The problem is that forecast errors are not equally costly in both directions: under-ordering a perishable loses the sale, over-ordering wastes the stock, and the two costs differ by product. Following the predict-then-optimise literature [26], [27], we instead train the forecaster on the cost of the order quantity it leads to, so that it learns to err in the cheaper direction. For replenishment this reduces to the classic newsvendor rule: order up to the demand level that is exceeded with probability $(1 - \tau)$, where

$$\tau_i = \frac{p_i}{p_i + h_i + \omega_i \Pr[\text{expiry}]} \tag{1}$$

Here p is the cost of losing one sale, h the cost of holding one unit for one period, ω the cost of writing off one unit, and Pr[expiry] the chance that a leftover unit expires before it sells. A high stockout cost pushes τ up (order more); a short shelf life pushes it down (order less). With stockout cost 9, holding cost 1, write-off cost 4 and a 50% expiry chance, $\tau = 9 / (9 + 1 + 2) = 0.75$, so the right order is the 75th percentile of the demand distribution, not its average.

One correction is needed. When a product is sold out, the sales record shows zero for that period even if customers wanted it. A model trained on those zeros learns that demand was low, orders less next time, and the product sells out again. We therefore mark every period in which the product was unavailable for part of the time, and the model treats those sales figures as a lower bound rather than the truth.

D. Five-layer framework

Fig. 1 shows the framework. Layer 1 collects the data: orders, stock levels with expiry dates, picking and rider logs, the catalogue, weather and events, and customer reviews with their images. Layer 2 turns these into features (time, location, product, customer, operations and trust) in a single shared feature store, built so that a model in training only ever sees data that would have been available at that moment. Layer 3 holds the models: demand forecasting, recommendation and substitution, routing and arrival-time estimation, inventory optimisation, and the proposed review-authenticity component. Layer 4 turns model outputs into a ranked list of suggested actions with explanations, and hosts the LLM-based assistant. Layer 5 is the human planner, who works from a queue of exceptions and whose approvals and rejections are stored.

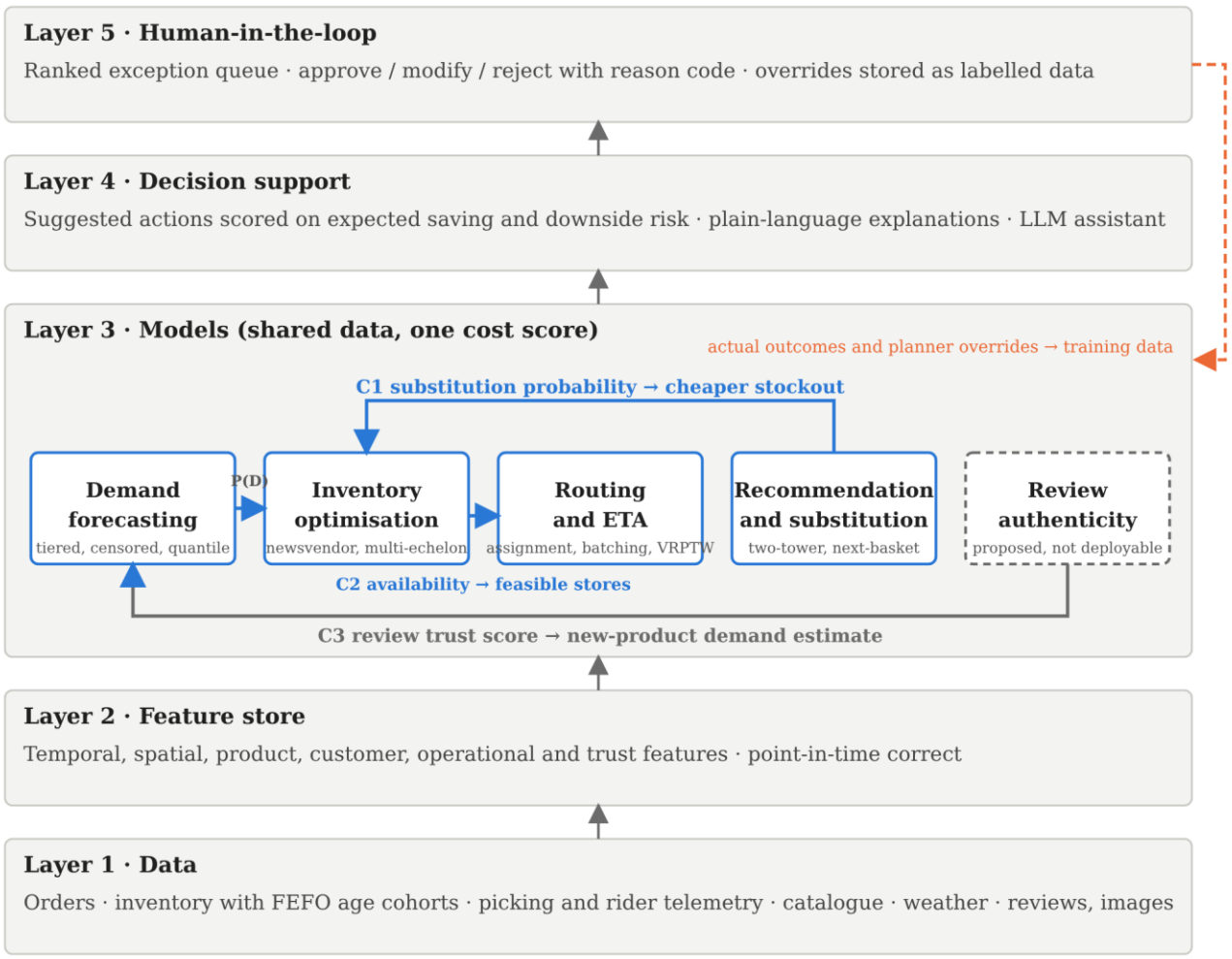


Fig. 1. The five-layer framework. Blue edges are the couplings C1–C3; the dashed orange edge is the path by which planner decisions and realised outcomes feed back into model training.

Two things separate this from a set of independent models. The first is the three couplings inside Layer 3: the forecaster passes a demand range, not a single number, to the inventory optimiser; the inventory optimiser passes projected stock to order assignment (C2); the recommender passes substitution probabilities that change the cost of a stockout (C1); and the authenticity component weights the reviews used for new-product forecasts (C3). The second is the return path from Layer 5 to Layer 3. What actually happened after each decision becomes the training signal for the next round, and every planner override, with its reason code, becomes a labelled example of a case the model got wrong.

E. Models and decision support

Demand forecasting uses a ladder of models so that every forecast can be traced to the method that produced it. Tier 0 holds simple baselines (same time last week, exponential smoothing, Prophet [14]) that any learned model must beat. Tier 1 is the workhorse: gradient-boosted trees [15], [16] trained on all product–store series at once, with product and store identity as inputs, and trained to predict a range of outcomes rather than a single number, because rule (1) needs a percentile. Tier 2 adds deep models [18], [17] for products with enough history to benefit. Tier 3 handles rarely-selling and brand-new items using intermittent-demand methods [21], reconciliation across the store–zone–city hierarchy [19], [20] and pretrained time-series models [22], [23]. The planner is told which tier produced each forecast.

Recommendation combines a retrieve-then-rank design [36] with a next-basket model built for repeat grocery purchases [38], always compared against the "what they bought last time" baseline [39]. A separate model estimates the substitution probability used in C1. Routing uses a learned travel-time model that outputs a range, so the platform can promise a delivery time it will meet nine times out of ten [34]; batching and sequencing are solved as a vehicle routing problem with time windows using OR-Tools [55], re-planned every few minutes as new orders arrive [31], [32], [33]. Inventory optimisation uses the forecaster's demand range directly with rule (1) [24]; reinforcement learning is tested as an option rather than assumed to be better [25].

The review-authenticity component combines three checks: a text classifier, treated as the weakest signal because AI-written text is now hard to tell from human text; an image check for AI-generated, edited or reused photos [52], [53], [54]; and a graph check on how reviewers, products and timing connect, which is where coordinated fake-review campaigns show up [46], [47]. The output is a recommendation to a human moderator with the evidence attached, never an automatic takedown, in line with IS 19000:2022 [11] and the CCPA dark-patterns rules [12].

Combining text and graph checks has a published precedent [51]; adding the image check does not, and the three together are untested, so this component is presented as proposed and not yet ready to deploy.

Every suggested action in Layer 4 is scored on how much it is expected to reduce the cost score and on how bad its worst realistic outcome is, so that a suggestion with a good average but a nasty downside is not ranked above a safer one. Actions are limited to a fixed list (reorder, move stock between stores, change a substitution rule, delist a product, change a dispatch or route, flag a capacity problem or a suspicious review). Each suggestion comes with a plain-language explanation: which inputs drove it (using SHAP values [56]), what would have to change for the suggestion to flip [57], and the estimated risk of a stockout, a late delivery or waste. The LLM-based assistant is split into small agents with fixed jobs (watch for exceptions, diagnose, propose options, check against policy, write the summary) [40], [41], [42], under three rules: every number comes from a verified calculation tool, never from the language model; every action that affects operations needs a human's approval; and every output follows a fixed, auditable format. Rejecting or changing a suggestion requires a reason code, and that record of where the model was judged wrong is the system's most valuable training data.

IV. DATA AND EXPERIMENTAL SETUP

A. Indian platform data

Table I summarises the disclosed operating position of the seven Indian platforms used to set the problem's parameters; three of them publish the operating metrics the simulator needs. The purpose is not ranking but fixing the values at which the problem is actually posed. Blinkit reports NOV, Swiggy GOV and Zepto net revenue, which are different measures, so they are never summed or averaged. Fig. 2 places the market-size estimates from Section I on one scale, and Fig. 3 shows network size by platform.

TABLE I. DISCLOSED OPERATING PARAMETERS OF INDIAN QUICK-COMMERCE PLATFORMS

Platform	Dark stores (as of)	Orders / AOV	Unit economics	Source
Blinkit (Eternal Ltd.)	2,443 (30 Jun 2026)	331M orders (Q1 FY27); net AOV ₹518	Adj. EBITDA +0.6% of NOV; inventory loss ₹308 crore (1.8% of NOV)	[7]
Swiggy Instamart	1,171 (30 Jun 2026)	114.5M orders (Q1 FY27); gross AOV ₹691; 1,089 orders/store/day	Contribution margin −0.2% of GOV; ~40% utilisation headroom	[8]
Zepto	1,139 (31 Mar 2026)	640M orders (FY26); ~2,140 orders/store/day (prospectus; see text)	Revenue ₹22,624 crore (+104% YoY); FY26 net loss ₹5,905 crore (+26% YoY)	[58]
Flipkart Minutes	~1,000 MFCs (Jun 2026)	Not disclosed	Not disclosed	[59]
BigBasket / BB Now	664+ (Jul 2026, third-party count)	Not disclosed	B2C revenue ₹8,223 crore (FY26)	[60], [61]
Amazon Now	500+ MFCs (Jun 2026)	Not disclosed	Not disclosed	[59]
JioMart	600+ (Sep 2025)	Grocery digital orders +116% YoY (Q1 FY27)	Not disclosed	[62], [63]

AOV = average order value; NOV = net order value; GOV = gross order value; MFC = micro-fulfilment centre; YoY = year on year. Blinkit's net AOV is derived from disclosed NOV and order count. "Not disclosed" means the operator does not publish the figure. Operators report different aggregates (NOV, GOV, net revenue), which are not summed or averaged in this paper.

Three observations shape the design. First, the operational problem rather than customer acquisition is now the binding constraint on profitability: Blinkit's single-quarter inventory loss exceeds its whole margin several times over [7]. Second, range size varies fourfold within one network, from roughly 80,000 products in Delhi NCR to roughly 20,000 beyond the top eight cities [1], so a single forecasting model for the whole country needs city tier as an input. Third, Swiggy publishes store-level throughput, 985 to 1,089 orders per store per day between Q1 FY26 and Q1 FY27 [8], which is the sector's best public series for calibrating an operations model. Zepto's 2,140 orders per store per day comes from its draft prospectus for Q4 FY26, does not match its own totals exactly (2,333,488 daily orders over 1,139 stores gives about 2,049), and is counted differently from Swiggy's figure, so the two are reported with these caveats rather than compared [58].

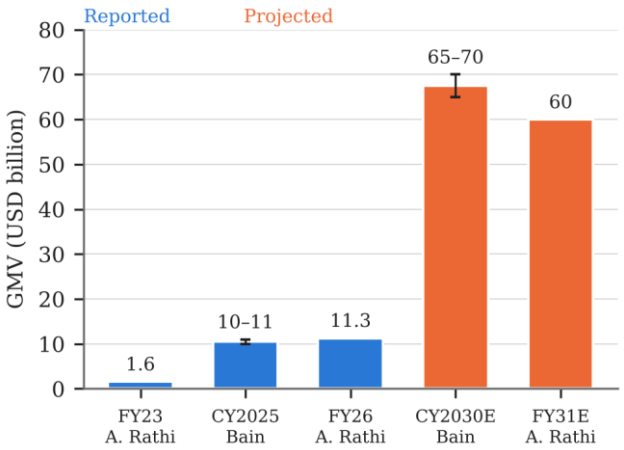


Fig. 2. Indian Q-commerce GMV estimates (USD billion). Bars show the midpoint and whiskers the published range. Sources: Anand Rathi [3] (fiscal years) and Bain–Flipkart [2] (calendar years); the two series use different year bases and are not spliced.

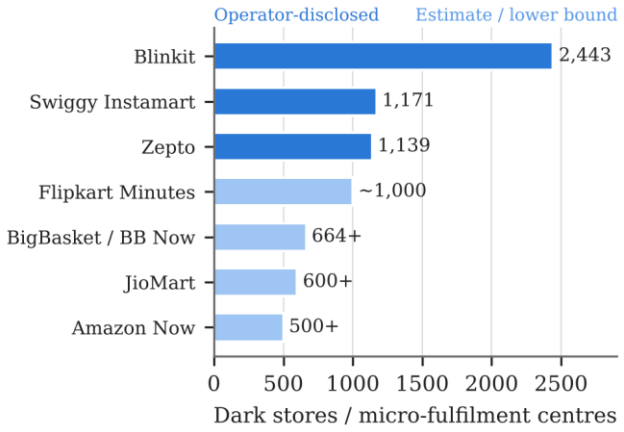


Fig. 3. Dark-store counts by platform at the dates in Table I. Dark bars are operator disclosures [7], [8], [58]; light bars are estimates or lower bounds from company statements and third-party counts.

B. Public datasets and simulator

For reproducible development we use public data. The Instacart Online Grocery Shopping Dataset 2017 (over three million orders from more than 200,000 users across about 50,000 products) is used for reorder prediction, next-basket recommendation and the basket statistics that drive the simulator's order generator [37]. M5 provides daily sales for a hierarchy of Walmart stores and products, used for the forecasting and rare-demand work [13]. For authenticity, the Amazon Reviews 2023 corpus and YelpChi [46] supply review text, images and the reviewer–product graph. None of these is Q-commerce data and none is Indian; Section VI treats this as the main limitation.

No public dataset captures dark-store operations, so we specify a discrete-event simulator with a configurable store layout, an order-arrival process that varies by time of day and weather, random picking times, riders who log in and out, travel times over a real road network, and expiry-aware inventory. Its structural parameters (number of stores, orders per store per day, range size by city tier, order value and margins) are set from the disclosures in Table I, with Swiggy's throughput series the single most useful input [8]. Parameters that public data cannot reach, such as pick rates, rider speeds and the mix of shelf lives, are stated as assumptions and varied to see how much the results depend on them.

C. Baselines, metrics and hypotheses

Each component is compared against a properly tuned baseline: a global LightGBM model for forecasting, the repeat-purchase model for recommendation [39], OR-Tools for routing [55], a fixed reorder-point rule for inventory, and single-signal detectors for authenticity. The baseline that matters most for the central claim is the siloed stack, in which each component is tuned on its own and the outputs are simply chained together. Metrics are reported at three levels: model level (forecast error and whether predicted ranges are reliable), operations level (fill rate, share of late orders, cost per order, waste rate and decision regret) and planner level (how often suggestions are accepted or overridden, and whether accepted suggestions turned out better than overridden ones). Training always uses earlier weeks and testing later weeks. Decision regret, the cost of the decision taken minus the cost of the best decision in hindsight, can only be computed in the simulator, where the best decision is known.

We state what we expect to see so that the evaluation can prove us wrong. H1: one gradient-boosted model trained across all products and stores beats one model per series on forecast error, with the biggest margin on rarely-selling items [13]. H2: training the forecaster on decision cost beats training it on accuracy when measured on operating cost, even though it scores worse on the usual accuracy metric; this is the central claim, and the fact that accuracy gets worse is what makes it a sharp test. H3: pretrained time-series models beat category averages for new products and stores in the first few weeks, after which models fitted on the product's own history overtake them [22], [23]. H4: forecasts with reliable ranges reach the same availability with less stock, so availability and waste improve together rather than one at the expense of the other. H5: the integrated stack beats the siloed stack on operating metrics, and the gain comes from the couplings; the test switches off C1, C2 and C3 one at a time, and if the integrated stack is no better than the siloed one in all three cases, the paper's thesis is wrong.

V. RESULTS AND DISCUSSION

A. Preliminary experiment

As a first check on H2 we compared the two training approaches on simulated rare-demand data, since no Q-commerce data is available to us. Two hundred product series were generated with demand that occurs only on some days and in small quantities, giving 11,000 test periods. Both models use the same inputs, the last eight periods of demand. The accuracy-trained model is a regression that predicts the average demand. The decision-trained model predicts the 75th percentile instead, which is what rule (1) prescribes for stockout cost 9, holding cost 1, write-off cost 4 and a 50% expiry chance. The decision-trained model is worse on the accuracy metric (RMSE 3.18 against 2.70) but better on the operating metric (inventory cost per period 12.05 against 13.44, a 10.3% reduction) and keeps more products in stock (fill rate 0.66 against 0.39). Fig. 4 shows the three metrics. This is the direction H2 predicts, but it is a simulated signal only: the rule is known to be right in theory, so the experiment confirms that the implementation works, not that the hypothesis holds on real data. The script and results file are supplied with the paper.

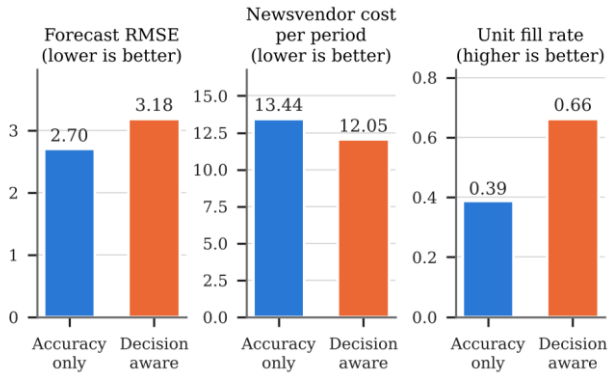


Fig. 4. Accuracy-trained versus decision-trained forecasting on simulated rare-demand data (200 series, 11,000 test periods, target percentile 0.75). The decision-trained model loses on RMSE and wins on cost and fill rate.

B. Where integration exists and where it does not

Table II records, for each research area, what the literature has achieved, what it has not, why, and the route

this framework takes. The same barrier appears in nearly every row: no public dataset couples local demand, inventory, availability and delivery, and platforms disclose almost nothing about what they run. The gap is one of evidence before it is one of algorithms.

TABLE II. INTEGRATION GAP ANALYSIS ACROSS THE SEVEN RESEARCH AREAS

Area	What exists	What does not	Why not	Route taken here
Demand forecasting	Hierarchical retail forecasting (M5); probabilistic deep models; zero-shot foundation models [13], [18], [22]	SKU-store-15-minute forecasting with availability censoring in Q-commerce	No public Q-commerce data with availability masks	Treat sold-out periods as lower bounds; testable on M5 with simulated stockouts
Decision-focused learning	SPO for linear and newsvendor problems [26], [27]	DFL coupled to stochastic routing and multi-echelon inventory	Non-differentiable combinatorial objective; no benchmark	Cost-aware training for replenishment only; routing keeps a standard forecast
Last-mile routing	Meal delivery with deadlines; courier dispatch theory; learned ETA [31], [33], [34]	Inventory availability as an input to order-to-store assignment	OR literature treats inventory as given; separate research communities	Availability projection constrains the feasible store set (C2)
Recommendation	Two-tower retrieval; next-basket; repeat-aware grocery models [36], [38]	Substitution acceptance feeding stockout cost in inventory	RecSys and OR do not publish in each other's venues	Stockout cost discounted by substitution probability (C1)
Agentic / LLM	Capability demos; LLM-to-solver interfaces [40], [41]	Measured operational outcomes in production	Technology under three years old; no platform has published outcomes	Bounded role; value stated as a hypothesis
Review authenticity	Text, graph and image detection separately; text + graph fusion [47], [51], [54]	Image-forensic + graph fusion; any review-image benchmark; any link from authenticity to cold-start	No labelled dataset exists; detector precision on real data still low	Architecture specified; component marked not deployable; enters only through C3
Unified DSS	Component DSSs in retail and logistics [28]	One Q-commerce DSS with a shared objective across the six areas above	No public data couples demand, inventory, availability and delivery; operators disclose little about AI deployment	Simulator set from disclosed figures; call for a public benchmark

The most telling result of the review is how little is public about AI deployment in Indian Q-commerce. After removing vendor marketing, the verifiable record across the seven platforms reduces to four items: a Swiggy engineering post on hierarchical demand forecasting with a time-series foundation model [64], a Swiggy customer lifetime-value model [65], Swiggy's Model Context Protocol announcement [66] and an ElasticRun statement on product-level forecasting for Q-commerce supply chains [67]. Which models each platform runs, how deeply they are integrated and what they measurably achieve is not disclosed, and that absence is why the academic literature on Q-commerce operations is thin. BCG's finding that about 74% of companies struggle to get value from AI, mostly for reasons of people and process rather than algorithms [68], suggests the gap is organisational as much as technical.

The framework answers these gaps with specific mechanisms: the three couplings replace siloed optimisation with shared state; rule (1) with the sold-out correction replaces accuracy-only training and breaks the loop in which stockouts cause under-forecasting and further stockouts; and plain-language explanations with a reason-coded override channel replace black-box outputs. The economic case follows from Section I: at margins near zero, small gains on holding, waste, delivery and picking have outsized effects.

VI. LIMITATIONS AND FUTURE SCOPE

This paper presents a framework and an evaluation plan, not results. The simulated experiment confirms that the implementation works, and every hypothesis in Section IV remains open on real data. The central claim, that integration and decision-aware training improve operating outcomes, is the kind of claim that is easy to argue and often fails to replicate.

The most serious limitation is that no public dataset covers the target domain. Instacart is multi-hour US grocery delivery with no inventory, delivery-time, rider or dark-store data; M5 is daily brick-and-mortar Walmart. Neither couples the demand, inventory, availability and delivery data on which the integration claims depend. The forecasting and recommendation components can be validated on public data; the operational and integration claims can only be tested in simulation or on a platform's private data, and a simulator built by the same authors as the method it evaluates is a weak instrument. Calibrating on Indian disclosures fixes the parameter problem but not the training-data problem: the demand patterns the models learn still come from US data, while Indian basket composition, price points, monsoon weather effects and gig-labour structure differ materially.

The authenticity component is the most speculative part of the framework. There is no benchmark for image authenticity in review content, the available synthetic-image datasets do not resemble the compressed, low-resolution photos in real reviews, the three-way combination is untested, and even the strongest published text-plus-graph detector reaches only about 34% precision on real Yelp data [51]. We would not deploy it on current evidence. Moderation also carries a cost that a pure optimisation view hides: a false positive silences a genuine customer. Likewise, treating rider utilisation purely as an efficiency number is incomplete, because fairness of assignment, earnings stability and safety belong in the objective too.

Future work centres on validation with an operating platform; a public review-authenticity benchmark for e-commerce; an anonymised or realistic synthetic Q-

commerce dataset that couples demand, inventory, availability, picking and delivery; and explicit modelling of emerging-market conditions. Building that shared evidence base would do more for the field than another model architecture.

VII. CONCLUSION

Quick commerce compresses the retail supply chain into a ten-minute window and in doing so turns periodic planning problems into continuous ones, at margins where individual decision errors show up in the accounts. This paper proposed a five-layer decision support framework that links demand forecasting, recommendation, inventory optimisation, last-mile routing and review-authenticity checks through three explicit couplings and one shared cost score. Forecasting models are trained on the cost of the orders they lead to, using product-specific costs that account for expiry, with a correction for sold-out periods. New products and stores are handled by a traceable ladder of fallback models, every suggestion comes with a plain-language explanation and a risk estimate, and a human planner stays at the point of decision with overrides captured as training data. The problem is grounded in disclosed figures from Indian platforms that crossed into single-digit profitability during 2026 against inventory losses several times their own margin. A simulated experiment shows the predicted trade-off between forecast accuracy and decision cost. We claim no results on real data. The central obstacle is evidence rather than method: there is no public dataset coupling local demand, inventory, availability and delivery, and until one exists, frameworks of this kind will keep being proposed more often than they are tested.

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